

CY 402 : Applied Materials Chemistry

Course Code : CY 402

Course Name : Applied Materials Chemistry

L-T-P-C : 3-0-0-3

Intended for : BS Chemical Sciences

Prerequisite : None

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- **Introduction to Materials chemistry:** Concepts of materials chemistry, Different types of materials, Sources and characteristics of traditional materials, Uses of materials, Approaches to producing new materials with new properties, Atomic level growth of solid material (crystalline and amorphous), Types of bonding in solids, Crystal structures. (6 Hours)
- **Polymers and Polymer based Materials:** Introduction to polymers, History and recent developments, Classification and nomenclature of polymers, Thermoplastics, Thermosets, Elastomers, Molecular weight, Polymer synthesis, Techniques of polymerization Conducting polymers. (9 Hours)
- **Nanomaterials:** Fundamentals of nanotechnology, Classification of nanomaterials, Synthesis of nanomaterials, Top down and bottom-up approach, Discovery and synthesis of quantum dots, Particle surface functionalization: electrostatic, steric and electrosteric stabilization, Toxicity. (9 Hours)
- **Biomaterials:** Introduction to biomaterials and its history, Classification, Properties of Biomaterials, biocompatibility and biodegradability, biopolymers, hydrogels, sealants and adhesives, Chemistry of dental materials . (9 Hours)
- **Functional materials, properties, and their applications:** Smart materials, Carbon materials, Energy materials, Optoelectronic materials, Catalysis, Environment, Agriculture, Biomedicine, Emerging materials, Thin films, Chemical vapour deposition (CVD), Atomic Layer deposition (ALD). (9 Hours)

Textbooks:

1. Harry R. Allcock, **Introduction to Materials Chemistry**, 2nd Edition, Wiley, 2019.
2. V. R. Gowarikar, N. V. Viswanathan, J. Sreedhar, **Polymer Science**, 3rd Edition, New Age International, Wiley, 2019.
3. Bikramjit Basu, **Biomaterials Science and Tissue Engineering: Principles and Methods**, Cambridge University Press, 2017.
4. M.D. Ventra, S. Evoy, J.R. Heflin Jr. (Eds.), **Introduction to Nanoscale Science and Technology**, Kluwer Academic Publishers.
5. Charles Stanley Gibson, **The Chemistry of Dental Materials**, Classic Reprint, 2018.



References:

1. R. J. Young and P. A. Lovell, **Introduction to Polymers**, CRC Press.
2. C. E. Carraher, **Polymer Chemistry**, CRC Press.
3. L. M. Liz-Marsan and P. V. Kamat, **Nanoscale Materials**, Kluwer Academic Publishers.
4. **Advanced Biomaterials: Fundamentals, Processing and Applications**, John Wiley & Sons, Inc. 2009.
5. Related journal articles
6. Alan J. Heeger, **Semiconducting and Metallic Polymers: The Fourth Generation of Polymeric Materials**, J. Phys. Chem. B, Vol. 105, No. 36, 2001
7. T. A. Skotheim (Editor), **Handbook of Conducting Polymers**, Dekker, 1986, vol. 1–2.
8. Bing Zhou, Scott Han, Robert Raja, and Gabor A. Somorjai, (Editors), **Nanotechnology in catalysis**, vol.3, Springer 2007
9. Charles P. Poole Jr and Frank J. Owens, **Introduction to Nanotechnology**, Wiley India student Edition, 2008
10. K.J. Klabunde and Ryan M. Richard, **Nanoscale Materials in chemistry**.
11. Guozhong Cao, **Nanostructured Materials**, Imperial College Press, 2004